BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Network Threats

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. How can more active users affect network performance?

2. How does a brute-force attack work?
   A) Tries many possible passwords
   B) Sends a trusted-looking message
   C) Floods a web server
   D) Rearranges file fragments

3. "A MAC address is the text domain name entered into a browser." Correct this. Explain why.

4. Data must reach another network in the right order. "TCP chooses fonts; IP draws images." Correct this. Explain each protocol's job.

Today's objectives

Explain the core ideas of network threats using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Malware
Malicious software such as ransomware, spyware or viruses.

Social engineering
Manipulating people into revealing information or bypassing safeguards.

Denial of service
Overwhelms a service to degrade or prevent access.

Unauthorised access
Guessing credentials or exploiting flaws may expose or change protected information.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Malware
Malicious software such as ransomware, spyware or viruses.

Social engineering
Manipulating people into revealing information or bypassing safeguards.

Denial of service
Overwhelms a service to degrade or prevent access.

Unauthorised access
Guessing credentials or exploiting flaws may expose or change protected information.

I Do - Worked model

A staff member receives a message asking for their password through an unfamiliar link. Identify the risk and response.

- Classify the attempt as phishing/social engineering.
- Explain that the link could harvest credentials.
- Do not enter details; verify independently via official channels.
- Report to school IT and follow account-security procedures.

We Do - Guided practice

1. An online service receives huge volumes of requests and slows down. Diagnose.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the line that increments an observed failed-login count.

```
attempts = 2
...
print(attempts)
```

You Do - Independent exam questions

Q2. Give the meaning of phishing. [2 marks]

Q3. Explain how ransomware can harm a school. [3 marks]

Q4. Name one precaution against social engineering. [2 marks]

Exit Ticket - True or False?

Phishing is a form of social engineering.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. Users may compete for finite bandwidth, reducing the data rate available to each.

2. A - Tries many possible passwords

3. A MAC address identifies a network interface on a LAN. It is usually six pairs of hexadecimal digits.

4. TCP supports reliable delivery in order. IP handles addressing and routing between networks. Each protocol has its own job; encryption does not prove a site is honest.

I Do - Explain the model

Classify the attempt as phishing/social engineering.

Explain that the link could harvest credentials.

Do not enter details; verify independently via official channels.

Report to school IT and follow account-security procedures.

We Do - Model answers

1. Distinguish attack evidence from a normal traffic spike.

Code First - Answer

```
attempts = 2
attempts = attempts + 1
print(attempts)
```

Expected output: 3

You Do - Self-mark

Q2: Deceptive communication designed to trick a user into disclosing data or taking unsafe actions.

Q3: Malware may encrypt/deny access to files, disrupt services and cause recovery costs.

Q4: Staff training, verifying requests independently or multifactor authentication.

GREEN PEN: Choose the hardest answer. Correct it on lined paper.